The irrationality exponents of $\log(2) + \sqrt{2}$ and $\sqrt{2}\log(2)$ are at most 4

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Abstract

We prove that each of the transcendental numbers

$$\log 2 + \sqrt{2}, \qquad \sqrt{2}\log 2, \qquad \frac{\log(3+2\sqrt{2})}{\sqrt{2}} + \sqrt{3}$$

has irrationality exponent at most 4. For each of these numbers x and each real $\nu > 4$, the proof gives an eventual lower bound $q^{-\nu} \le |x - p/q|$, uniformly in the integer numerator and positive integer denominator; the fractions need not be reduced. We prove all three bounds using the same determinant argument. Working in a number field of degree four gives four conjugate versions of the determinant. Two satisfy an exponential identity that allows a strong analytic bound; the other two are controlled by simpler estimates on their coefficients. These four values come from the same cleared algebraic-integer determinant. Its norm supplies the arithmetic lower bound. A Lean formalization accompanies the proof.

1 Introduction and main results

Adding an algebraic number to a logarithm, or multiplying the logarithm by one, changes the rational approximation problem. An approximation to $\log 2 + \sqrt{2}$ becomes an approximation to $\log 2$ by an element of $\mathbb{Q}(\sqrt{2})$, rather than by a rational number. The same issue appears for $\sqrt{2}\log 2$. We construct the interpolation determinant over a number field to obtain the following bounds. All three numbers are transcendental by the Hermite–Lindemann theorem [1, Section 2]; our concern here is the quality of their rational approximations.

Throughout the paper, log denotes the natural real logarithm on positive arguments, and square roots denote positive real square roots. For an irrational real number x, its irrationality exponent $\mu(x)$ is the supremum of the real numbers ν for which $0 < |x - p/q| < q^{-\nu}$ holds for infinitely many reduced fractions p/q with positive denominator. Section 2 gives the conventions used in the proof.

Theorem A (An algebraic shift)**.** *The number* $x_1 = \log 2 + \sqrt{2}$ *satisfies*

$$\mu(\log 2 + \sqrt{2}) \le 4.$$

Theorem B (An algebraic multiple)**.** *The number* $x_2 = \sqrt{2}\log 2$ *satisfies*

$$\mu(\sqrt{2}\log 2) \le 4.$$

Theorem C (A shifted quadratic period)**.** *The number*

$$x_3 = \frac{\log(3+2\sqrt{2})}{\sqrt{2}} + \sqrt{3}$$

satisfies $\mu(x_3) \le 4$.

For each of the three numbers, we prove the more precise statement

$$\forall \nu > 4 \quad \exists Q \in \mathbb{N},\ Q \ge 2 : \quad \forall p \in \mathbb{Z}\ \forall q \in \mathbb{N}, \quad q \ge Q \implies q^{-\nu} \le \left| x - \frac{p}{q} \right|. \tag{1}$$

The threshold depends on x and ν, but not on the numerator or denominator. There is no coprimality condition in (1).

1.1 Comparison with earlier work

The first two bounds can be compared with Theorem 1 of Luchin and Salikhov [2] (2018) on approximation to $\log 2$ by elements of $\mathbb{Q}(\sqrt{2})$. Their theorem implies

$$\mu(\log 2 + \sqrt{2}) \le 7.98101, \qquad \mu(\sqrt{2}\log 2) \le 7.98101.$$

Theorems A and B improve these bounds to 4. The comparison follows by writing

$$\left| \log 2 + \sqrt{2} - \frac{p}{q} \right| = \left| \log 2 - \frac{p - q\sqrt{2}}{q} \right|$$

and

$$\left| \sqrt{2}\log 2 - \frac{p}{q} \right| = \sqrt{2} \left| \log 2 - \frac{p}{q\sqrt{2}} \right|.$$

In both substitutions, their coefficient height is $\max(|p|, q)$, which is comparable to q whenever the approximation error is less than one. The fixed factors are absorbed by an arbitrarily small increase in the exponent.

1.2 Proof overview

We extend the logarithmic determinant argument to algebraic shifts and multiples by estimating one algebraic determinant at all four embeddings of a number field. At two embeddings σ, integral elements a, u, v satisfy

$$\exp\big(\sigma(u)(x - \sigma(v))\big) = \sigma(a).$$

These are the embeddings where good rational approximations to x produce small translation errors. At the other embeddings we need only a bound on the matrix coefficients. Interpolation selects a nonzero minor at one positive real embedding. We clear that minor over the ring of integers, keep its column selection fixed, and take its norm.

The factor four in the final exponent comes from this comparison. There are four copies of the arithmetic clearing cost and two copies of the full analytic saving. After division by two, the leading arithmetic cost is $2(1 - b_H)$, where b_H is the mean transverse row weight. The analytic estimate is stronger than this arithmetic bound when $\nu > 4$ and the parameters are chosen suitably. The coefficient estimates at the other embeddings contribute additional terms. We

make those terms small by choosing the dimension and approximation denominators first, then let the determinant height grow.

Our starting point is the weighted logarithmic interpolation and collision framework in OpenAI's `PiExponent` library [3], together with the rational-logarithm and quadratic-logarithm adaptations [4, 5]. The present argument uses those mathematical inputs, including their algebraic geometry and complex analysis. Sections 5 and 7 state the interpolation and collision theorems used below.

2 Approximation conventions

For an irrational real number x, let

$$\mathcal{A}(x) = \left\{ \nu > 0 : \begin{array}{l} \text{infinitely many distinct reduced fractions } p/q,\ q \geq 2, \\ \text{satisfy } 0 < |x - p/q| < q^{-\nu} \end{array} \right\}.$$

We use $\mu(x) = \sup \mathcal{A}(x)$, allowing the value $+\infty$. Dirichlet approximation gives $2 \in \mathcal{A}(x)$ for irrational x.

Lemma 2.1. *If a real number x satisfies (1), then x is irrational and $2 \leq \mu(x) \leq 4$.*

Proof. Suppose $x = r/s$, with $r \in \mathbb{Z}$ and s a positive integer. Apply (1) with $\nu = 5$. For an integer N large enough that $Ns \geq Q$, put $p = Nr$ and $q = Ns$. The approximation error is zero, contradicting $q^{-5} > 0$.

Now fix $\nu > 4$. Every reduced fraction counted in $\mathcal{A}(x)$ at exponent ν has $q < Q$. Its error is less than one, hence $|p| < q(|x| + 1)$. There are only finitely many such denominators and numerators. Therefore no $\nu > 4$ belongs to $\mathcal{A}(x)$, which proves the upper bound. Dirichlet's theorem gives the lower bound. □

We will prove (1) by excluding an unbounded supply of denominators for the opposite estimate. More precisely, for each fixed $\nu > 4$ we rule out

$$\forall Q \in \mathbb{N} \quad \exists p \in \mathbb{Z}\ \exists q \in \mathbb{N}, \quad q \geq Q, \qquad \left| x - \frac{p}{q} \right| \leq q^{-\nu}. \tag{2}$$

Its negation gives a threshold beyond which the error is strictly greater than $q^{-\nu}$. Increasing that threshold to at least two and weakening the inequality gives (1).

3 The common degree-four theorem

For a number field F, write $\mathcal{O}_F$ for its ring of integers and $\mathrm{Emb}(F) = \mathrm{Hom}_{\mathbb{Q}\text{-alg}}(F, \mathbb{C})$. Embeddings are counted as maps. In particular, two distinct embeddings can have the same value on a, u, v.

Theorem 3.1 (Two compatible embeddings)**.** *Let F be a number field with $[F : \mathbb{Q}] = 4$, and let $a, u, v \in \mathcal{O}_F$. Suppose there is an embedding σ_0 such that $\sigma_0(a) = b_0$ is real, $b_0 > 0$, and $b_0 \neq 1$. Let $x \in \mathbb{R} \setminus \{0\}$. Suppose a set $G \subset \mathrm{Emb}(F)$ of exactly two embeddings satisfies*

$$\exp(\sigma(u)(x - \sigma(v))) = \sigma(a), \qquad |\sigma(u)| \leq 2 \qquad (\sigma \in G). \tag{3}$$

Then x satisfies (1), and consequently is irrational with $\mu(x) \leq 4$.

The distinguished embedding σ_0 is used for interpolation. It need not belong to G. The theorem imposes no exponential compatibility at the two remaining embeddings, and they need not be real. Complex absolute values are used throughout the common argument. We call the members of G the *good embeddings*, meaning only that (3) holds there.

3.1 Fixed bounds for the embeddings

Fix $\nu > 4$ and assume (2). Before choosing any rational approximations, put

$$\kappa = \sum_{\sigma \in \mathrm{Emb}(F)} \log \max(1, |\sigma(a)|), \tag{4}$$

$$R = \sum_{\sigma \in \mathrm{Emb}(F)} |\sigma(u)|(|x| + 1 + |\sigma(v)|), \qquad R' = 2R + 2,$$

$$\rho = 100 \max\left(1, |x|, \sum_{\sigma \in \mathrm{Emb}(F)} |\sigma(u)(x - \sigma(v))|\right).$$

All summands are nonnegative. Thus $\kappa, R \geq 0$, $\rho \geq 100$, $100|x| \leq \rho$, and, separately for every embedding,

$$\begin{aligned} \log \max(1, |\sigma(a)|) &\leq \kappa, \\ |\sigma(u)|(|x| + 1 + |\sigma(v)|) &\leq R, \\ 100|\sigma(u)(x - \sigma(v))| &\leq \rho. \end{aligned} \tag{5}$$

These bounds keep the later choices uniform over the embeddings. The enlargement from R to R' absorbs the fixed multiplier in the affine approximation error.

4 Choosing the parameters

The proof proceeds as follows. This section chooses parameters with the two strict margins in (19). Section 5 then selects a nonzero minor, and Section 6 clears its denominators to obtain the norm inequality. The same minor is estimated at the two good embeddings in Section 7 and at the other embeddings in Section 8. Section 9 combines these estimates with the norm inequality; the two margins make them incompatible.

The following table collects the notation used in these steps. The last column points to the definitions.

Symbols	Meaning	Definition
κ, R, R', ρ	Fixed bounds, uniform over the embeddings	(4)
$\theta, A, B, C, \eta;\ g$	Shape parameters and the positive approximation margin	(7)–(8)
F_0, ζ	Truncation parameter and geometric slack	Section 4
$m, K;\ w_0, v_0$	Number of approximations, number of centers, and weights in the first coordinate	(9)
$p_i/q_i,\ w_i, w_*$	Chosen approximations, their logarithmic weights, and the smallest weight	(14)
$E, U, T, J;\ V = T + J$	Clearing, coarse, translation, and holomorphic error terms	(17)
$H, M;\ \gamma, \beta$	Matrix height, row count, and column and row multi-indices	(22)
$\Delta_\sigma, \mathscr{C}_H, Z_H$	Selected minor at σ, clearing factor, and algebraic integer	(27), (28)
b_H	Mean transverse row weight, divided by H	(30)
α, d, k	Assigned transverse index, tail powers, and tail coefficient order in a translation term	(39)
N_H, C_H, C_∞	Low-index count, collision rate, and its limit as $H \to \infty$	(43), (12)

There are two quite different limiting operations in the proof. First we choose a dimension and a finite set of good rational approximations. Then we hold them fixed and increase the matrix height H. Keeping that order matters: a term such as $1/w_*$, where w_* is the smallest approximation weight, does not tend to zero with H.

Set

$$\Lambda = \log 4 + 4, \qquad c = \frac{\log 2}{4}. \tag{6}$$

The first is an admissible constant for the least-common-multiple estimate; the second is the collision constant. Choose rational shape parameters

$$0 < \theta < A < B < 1, \quad C > 1, \quad B < C^{-1}, \quad B < C\theta < 1, \tag{7}$$

and a real $0 < \eta < 1$, so that

$$g = \nu(A(1-\eta) - \theta) - 2(1-\theta) > 0. \tag{8}$$

Here is a direct way to see that such a shape exists. Choose rational $t < 1$ sufficiently close to one that $\nu t > 2(1+t)$, and put $\theta = t^2$. At $A = t$ and $\eta = 0$, the expression in (8) is $(1-t)(\nu t - 2(1+t)) > 0$. By continuity, we may choose rational $\theta < A < B < t$ sufficiently close to t, followed by a small positive η, while keeping the gap positive. Choose rational C in the nonempty interval $(B/\theta, 1/B)$. Then (7) holds, including $C\theta < \theta/B < 1$.

The remaining choices have two purposes. The terms from denominator clearing, coarse coefficient bounds, and row translation must have sum less than g. The collision estimate must also exceed the arithmetic bound when the translated indices have small total weight. These are the two inequalities in (19) below. We write their error terms as E for clearing, U for the coarse bound, and $V = T + J$ for translation and holomorphic growth. Increasing the dimension makes the limiting collision rate large; increasing the approximation weights then makes the remaining errors small.

Choose $0 < \epsilon < g$ with $\epsilon \le 1/2$, and then choose $F_0 > 2/\theta$ such that $\nu/F_0 < \epsilon/3$. For a sufficiently large integer $m \ge 1$, set

$$K = \lfloor C^m \rfloor, \qquad w_0 = B^{-m}, \qquad v_0 = 2K\theta^m w_0. \tag{9}$$

Here $K \ge 1$, the weights w_0, v_0 are positive rationals, and

$$K\frac{w_0}{v_0}\theta^m = \frac{1}{2}, \qquad K\theta^m < 1. \tag{10}$$

We choose m to make

$$\frac{2\Lambda F_0 m + 3\log 2}{v_0} + \frac{(\rho+\kappa)K + \log(3/2)}{w_0} < \frac{\epsilon}{3} \tag{11}$$

and simultaneously

$$C_\infty := \frac{c\eta^2 K\theta^m}{(m+1)v_0 A^m} > 3. \tag{12}$$

With the shape fixed, $K/w_0 \le (CB)^m \to 0$, while v_0 grows exponentially because $C\theta/B > 1$. For example, once $C^m \ge 2$, $K \ge C^m/2$, so $v_0 \ge (C\theta/B)^m$. Thus even $m/v_0 \to 0$. On the other hand, substitution from (9) gives the exact identity

$$C_\infty = \frac{c\eta^2}{2(m+1)}\left(\frac{B}{A}\right)^m,$$

which tends to infinity.

Choose a small positive rational ζ such that

$$\frac{(1+3\zeta)^{m+1}}{2} < 1, \qquad (1+3\zeta)^m K\theta^m < 1, \qquad (1+\zeta)\theta < 1. \tag{13}$$

All three inequalities follow by taking ζ close enough to zero, since their zero-slack versions are strict.

4.1 Successive approximation weights

The hypothesis (2) supplies a sequence of approximations whose denominators can be required to exceed any previously specified threshold. Choose its first m members p_i/q_i, $1 \le i \le m$, with

$$q_i \ge 2, \quad p_i \ne 0, \quad |x - p_i/q_i| \le q_i^{-\nu}, \qquad w_i = \lceil \log q_i \rceil \ge 1. \tag{14}$$

The condition $p_i \ne 0$ is available because $x \ne 0$: zero numerators cannot satisfy the approximation inequality once q_i is sufficiently large. Let $w_* = \min_i w_i > 0$.

The weights must also be sufficiently separated for interpolation. Here is the precise finite condition we impose. Set

$$d = (w_0, w_1, \dots, w_m), \qquad e = (v_0, w_1/\theta, \dots, w_m/\theta),$$

and

$$D_m(\zeta) = 2(m+2)^{m+2}\left(1 + \frac{m+2}{\zeta}\right)^{m+2}. \tag{15}$$

For equal-cardinality subsets $P, Q \subseteq \{0, \dots, m\}$, whenever $i > 0$ belongs to $P \setminus Q$ and membership in P, Q agrees at every index above i, require

$$D_m(\zeta) \prod_{j \in Q} e_j < \prod_{j \in P} d_j. \tag{16}$$

After common indices are cancelled, the largest remaining index is on the right. Its weight can dominate the finitely many products of earlier weights. Successively increasing the next denominator therefore achieves these inequalities.

Finally choose the initial denominator threshold large enough to make the remaining inverse-weight errors small. Define

$$E = \frac{\Lambda F_0 m}{v_0} + \Lambda \sum_{i=1}^{m} \frac{1}{w_i} + \frac{\theta}{w_*}, \tag{17}$$

$$U(S) = \frac{\log 2}{v_0} + \frac{\kappa K + \log(3/2)}{w_0} + \frac{\log(2(KS+2))}{w_*} \quad (S \ge 0),$$

$$T = \frac{\nu}{F_0} + \frac{\log 2}{v_0} + \frac{\log 4 + \log(2K) + \nu}{w_*},$$

$$J = \frac{\rho K}{w_0} + \frac{\log 2}{v_0} + \frac{\log(200K) + \log(\rho/100)}{w_*}, \qquad V = T + J.$$

The part of $2E + U(R') + V$ independent of w_i is ν/F_0 plus the left side of (11). The rest is

$$2\Lambda \sum_i \frac{1}{w_i} + \frac{2\theta + \log(2(KR'+2)) + \log 4 + \log(2K) + \nu + \log(200K) + \log(\rho/100)}{w_*}. \tag{18}$$

Since $\sum_i 1/w_i \le m/w_*$, an arbitrarily large initial weight makes (18) less than $\epsilon/3$, without interfering with later separation. We have proved the two margins

$$2E + U(R') + V < g, \qquad 2 + 2E + U(R') + V < C_\infty. \tag{19}$$

For the second inequality, the error sum is less than $\epsilon \le 1/2$, whereas $C_\infty > 3$.

The order of choice is now complete: shape, F_0, dimension, slack, initial approximation weight, successive approximations. From this point on, every quantity in (17) is fixed. Only H, the matrix indices, and the selected minor will vary.

5 Weighted interpolation and the matrix

We use the weighted logarithmic interpolation theorem from [4]. A polynomial in $Y, X_1, \dots, X_m$ has weighted degree at most H, with degree weights d_i, if every monomial $Y^h X^\gamma$ in its support satisfies $d_0 h + \sum_i d_i \gamma_i \le H$. A logarithmic jet packet with weights e_i consists of the coefficients of $t^s Z^\beta$ with $e_0 s + \sum_i e_i \beta_i < H$, after the formal substitution

$$Y = Y_j(1+t), \qquad X_i = c_{ji} + \log(1+t) + Z_i. \tag{20}$$

Here the logarithm is the formal power series at zero.

Proposition 5.1 (Weighted logarithmic interpolation)**.** *Let d_i, e_i be positive rational weights, $\zeta > 0$ rational, $Y_0, \dots, Y_{K-1}$ distinct nonzero complex numbers, and c_{ji} arbitrary complex numbers. Suppose*

$$K(1+3\zeta)^{m+1} \frac{\prod_{i=0}^m d_i}{\prod_{i=0}^m e_i} < 1, \qquad (1+\zeta)d_i < e_i \quad (1 \le i \le m), \tag{21}$$

and the separated-product condition (16) holds for these weights. There is a positive rational scale $\mathcal{R}$ such that, for every sufficiently large integer n, the map from polynomials of weighted degree at most $n\mathcal{R}$ to their logarithmic jet packets of weight strictly below $n\mathcal{R}$ at all K centers is surjective.

This interpolation theorem is proved in [4] using ampleness of an interpolation bundle and eventual restriction to jets, based on the weighted geometric results in [3]. The additive centers in (20) are unrestricted.

5.1 The finite matrix

For $H > 0$, define the column and row sets

$$\begin{aligned} \mathcal{C}_H &= \{(h,\gamma) \in \mathbb{N} \times \mathbb{N}^m : w_0 h + |\gamma|_w \le H\}, \\ \mathcal{R}_H &= \{(j,s,\beta) : 0 \le j < K, s \in \mathbb{N},\ \beta \in \mathbb{N}^m, v_0 s + |\beta|_w/\theta < H\}, \end{aligned} \tag{22}$$

where $|\alpha|_w = \sum_i w_i \alpha_i$, and put $M = M(H) = \#\mathcal{R}_H$. This is a positive integer because $(0,0,0)$ is a row. Set

$$T_i = \left\lceil \frac{F_0 w_i}{v_0} \right\rceil, \qquad L_i(t) = \sum_{1 \le k < T_i} \frac{(-1)^{k+1}}{k} t^k. \tag{23}$$

For an embedding σ, write

$$z_\sigma = \sigma(a), \qquad r_{\sigma i} = \sigma(u)\left(\frac{p_i}{q_i} - \sigma(v)\right). \tag{24}$$

Our matrix, with rows and columns from (22), is

$$\mathcal{M}_\sigma(H)_{(j,s,\beta),(h,\gamma)} = z_\sigma^{jh}[t^s Z^\beta]\,(1+t)^h \prod_{i=1}^{m} (Z_i + jr_{\sigma i} + L_i(t))^{\gamma_i}. \tag{25}$$

Lemma 5.2. *The matrix $\mathcal{M}_{\sigma_0}(H)$ has full row rank for a cofinal set of heights $H > 0$.*

Proof. Apply Proposition 5.1 with the weights d, e chosen above, multiplicative centers $Y_j = b_0^j$, and additive centers $c_{ji} = jr_{\sigma_0 i}$. Since $b_0 > 0$ and $b_0 \neq 1$, these multiplicative centers are distinct and nonzero. The volume ratio is $1/2$ by (10), the coordinate inequalities follow from $(1+\zeta)\theta < 1$, and separation is (16).

To pass to (25), let $\tau_i(t) = \log(1+t) - L_i(t)$. It has order at least T_i, and

$$v_0 T_i \geq F_0 w_i > w_i/\theta.$$

Thus $Z_i \mapsto Z_i + \tau_i(t)$ and its inverse preserve the weighted filtration with weights $v_0, w_i/\theta$. They induce inverse linear maps on the finite jet packet spaces. Applying this substitution to the truncated expression recovers the full logarithmic expression. Consequently the full and truncated packet maps are surjective at exactly the same heights.

In the monomial bases, the truncated packet map has precisely the entries (25). Proposition 5.1 therefore gives full row rank for all sufficiently large heights of the form $H = n\mathcal{R}$. Since $\mathcal{R} > 0$, these heights are cofinal. □

At each such height, choose a nonzero full-row minor and denote its column selection by s_H. Use this *same* column selection at every embedding and let

$$\Delta_\sigma = \det(\mathcal{M}_\sigma(H)|_{s_H}). \tag{26}$$

At this stage only $\Delta_{\sigma_0} \neq 0$ is known. The common integral lift below will show that all four determinants are nonzero.

6 Clearing the selected minor

Let $e_i^H = \lfloor H/w_i \rfloor$ and $D_i = \operatorname{lcm}(1, \ldots, T_i)$. Every selected column satisfies $\gamma_i \leq e_i^H$. Multiply row (j, s, β) by $\prod_i D_i^{e_i^H} q_i^{-\beta_i}$, and selected column (h, γ) by $\prod_i q_i^{\gamma_i}$. Their total determinant factor is the positive rational number

$$\mathscr{C}_H = \left(\prod_i D_i^{e_i^H}\right)^M \left(\prod_{r \in \mathcal{R}_H} \prod_i q_i^{-\beta_{ri}}\right) \left(\prod_{c \in s_H} \prod_i q_i^{\gamma_{ci}}\right). \tag{27}$$

This factor need not be an integer. The assertion we need is that the scaled entries, and therefore their determinant, are algebraic integers.

Lemma 6.1 (A common integral lift)**.** *There is an element $Z_H \in \mathcal{O}_F \setminus \{0\}$ such that*

$$\sigma(Z_H) = \mathscr{C}_H \Delta_\sigma \qquad \text{for every } \sigma \in \operatorname{Emb}(F). \tag{28}$$

In particular, $\Delta_\sigma \neq 0$ at every embedding.

Proof. Define

$$n_i = u(p_i - q_i v) \in \mathcal{O}_F.$$

Then $r_{\sigma i} = \sigma(n_i)/q_i$. If some $\beta_i > \gamma_i$, the entry in (25) is zero. In the remaining case, extracting Z^β gives the factor $\prod_i \binom{\gamma_i}{\beta_i}$ and powers with exponents $d_i' = \gamma_i - \beta_i \geq 0$. After the row and column factors have been applied, the entry is the image under σ of

$$a^{jh} \left(\prod_i \binom{\gamma_i}{\beta_i} \right) [t^s] \, (1+t)^h \prod_i D_i^{e_i^H} \left(j n_i + q_i L_i(t) \right)^{d_i'}. \tag{29}$$

Each polynomial $D_i(j n_i + q_i L_i(t))$ has coefficients in $\mathcal{O}_F$, because D_i clears every denominator occurring in L_i. Also $d_i' \leq \gamma_i \leq e_i^H$, so

$$D_i^{e_i^H} \left(j n_i + q_i L_i \right)^{d_i'} = D_i^{e_i^H - d_i'} \left(D_i (j n_i + q_i L_i) \right)^{d_i'}$$

is integral coefficientwise. The other factors in (29) are integral as well.

Use these entries to form one square matrix over $\mathcal{O}_F$, and let Z_H be its determinant. Determinants commute with embeddings, which proves (28). Since $\mathscr{C}_H > 0$ and $\Delta_{\sigma_0} \neq 0$, we have $Z_H \neq 0$. Every embedding of a field is injective, so $\sigma(Z_H) \neq 0$ for every σ. Equation (28) then gives the remaining assertion. □

Define the normalized mean transverse row weight by

$$b_H = \frac{1}{MH} \sum_{r \in \mathcal{R}_H} |\beta_r|_w. \tag{30}$$

The row condition gives $0 \leq b_H \leq \theta$.

Lemma 6.2 (Clearing cost)**.** *For every $H > 0$ and selected full-row minor,*

$$\log \mathscr{C}_H \leq MH(1 - b_H + E). \tag{31}$$

Proof. The elementary least-common-multiple estimate from [3] is $\log D_i \leq \Lambda T_i$. Since $T_i \leq F_0 w_i / v_0 + 1$ and $e_i^H \leq H/w_i$,

$$\sum_i e_i^H \log D_i \leq H \left(\frac{\Lambda F_0 m}{v_0} + \Lambda \sum_i \frac{1}{w_i} \right).$$

For the column factors, $\log q_i \leq w_i$ and the column degree bound imply $\sum_i \gamma_{ci} \log q_i \leq H$ for each selected column. For the inverse row factors, $\log q_i \geq w_i - 1$ gives

$$-\sum_{r,i} \beta_{ri} \log q_i \leq -MHb_H + \sum_{r,i} \beta_{ri} \leq -MHb_H + \frac{MH\theta}{w_*}.$$

Adding these three estimates in the logarithm of (27) proves the claim. □

The inverse row factors give the term $-b_H$, which matches the transverse weight in the analytic estimate.

6.1 The norm inequality

The nonzero algebraic integer Z_H has nonzero integer norm. Thus

$$1 \le |N_{F/\mathbb{Q}}(Z_H)| = \prod_{\sigma \in \mathrm{Emb}(F)} |\sigma(Z_H)|,$$

and (28) yields

$$0 \le 4 \log \mathscr{C}_H + \sum_{\sigma \in \mathrm{Emb}(F)} \log |\Delta_\sigma|. \tag{32}$$

Lemma 6.1 gives $\Delta_\sigma \neq 0$ for every σ.

Lemma 6.3 (Normalized comparison)**.** *Suppose* $\log |\Delta_\sigma|/(MH) \le B_H$ *at the two good embeddings and* $\log |\Delta_\sigma|/(MH) \le D_H$ *at the two remaining embeddings. Then*

$$-2(1 - b_H) - 2E - D_H \le B_H. \tag{33}$$

Proof. Insert Lemma 6.2 and the four determinant estimates into (32). The resulting upper bound on its right side is $4MH(1 - b_H + E) + 2MHB_H + 2MHD_H$, and is nonnegative. Divide by $2MH > 0$ and rearrange. □

7 The analytic estimates

Fix a good embedding σ and write

$$\omega = \sigma(u)(x - \sigma(v)), \qquad z = z_\sigma.$$

Then $e^\omega = z$ and $100|\omega| \le \rho$. The translated centers satisfy

$$\begin{aligned} r_{\sigma i} - \omega &= \sigma(u)(p_i/q_i - x), \\ |j(r_{\sigma i} - \omega)| &\le \exp\bigl(\log(2K) + \nu + \log 2 - \nu w_i\bigr), \qquad 0 \le j < K. \end{aligned} \tag{34}$$

Indeed, $|\sigma(u)| \le 2$, the approximation error is at most $q_i^{-\nu}$, and $\log q_i \ge w_i - 1$. The slightly weaker bound above matches the constants in [4]; the multiplier u contributes the additional $\log 2$, which is absorbed after combining the embeddings.

7.1 The collision input

For nonnegative integers h, multi-indices γ, α, and an integer j, consider the formal coefficient

$$e^{j\omega h}[t^\ell Z^\alpha](1+t)^h \prod_i (Z_i + j\omega + \log(1+t))^{\gamma_i}.$$

These are the entries to which the collision estimate applies. They are related to the entire functions

$$f_{h,\gamma,\alpha}(z) = e^{hz} \left(\prod_i \binom{\gamma_i}{\alpha_i} \right) z^{\sum_i (\gamma_i - \alpha_i)} \quad (\alpha \le \gamma), \tag{35}$$

and are zero when $\alpha \not\le \gamma$. The logarithmic coordinate is $z = j\omega + \log(1+t)$. Repeated transverse indices cause the corresponding rows to use the same family of entire functions, which is the source of the collision saving.

Proposition 7.1 (Scaled collision estimate)**.** *Let a square matrix have $M \geq 1$ rows indexed by r and columns indexed by c. Assign a transverse multi-index α_r, an integer $0 \leq j_r < K$, and an order $\ell_r \geq 0$ to each row; assign (h_c, γ_c) with $w_0 h_c + |\gamma_c|_w \leq H$ to each column. Suppose*

$$H, w_0, v_0, w_* > 0, \quad w_i \geq w_*, \quad \rho \geq 100, \quad 100|\omega| \leq \rho, \quad \sum_r \ell_r \leq MH/v_0.$$

Let $n_\alpha = \#\{r : \alpha_r = \alpha\}$, and let B be the matrix with entries

$$B_{rc} = e^{j_r \omega h_c} [t^{\ell_r} Z^{\alpha_r}] (1+t)^{h_c} \prod_i (Z_i + j_r \omega + \log(1+t))^{\gamma_{ci}}. \tag{36}$$

Then

$$|\det B| \leq \exp\left(-c \sum_\alpha n_\alpha^2 + MH(J + r_{\mathrm{coll}}(M, H)) \right), \tag{37}$$

where J is given in (17) and

$$r_{\mathrm{coll}}(M, H) = \frac{\log M + c - \log(1 - 2^{-1/2})}{H}. \tag{38}$$

This is the scaled form of the holomorphic collision estimate in [3, 4]. Its analytic domain is a disk of radius ρK. The centers lie well inside this disk: $|j\omega| + 3/4 \leq \rho K/2$. On its boundary, the weighted degree of the columns bounds the entire functions in (35) by

$$\exp\left(H \left(\frac{\rho K}{w_0} + \frac{\log(2\rho K)}{w_*} \right) \right).$$

The derivative-order budget supplies $\log 2/v_0$, and the identity $\log(2\rho K) = \log(200K) + \log(\rho/100)$ gives exactly J. The remaining collision calculation gives the coefficient $c = \log 2/4$ and remainder (38).

7.2 Finite row translation

For each row $r = (j, s, \beta)$, write

$$Z_i + j r_{\sigma i} + L_i(t) = Z_i + j\omega + \log(1+t) + j(r_{\sigma i} - \omega) - \tau_i(t).$$

Put $\varepsilon_{ri} = j(r_{\sigma i} - \omega)$. A translation term for this row is specified by

$$\alpha \in \mathbb{N}^m, \quad \alpha \geq \beta, \quad |\alpha|_w \leq H, \qquad 0 \leq d_i \leq \alpha_i - \beta_i, \qquad 0 \leq k \leq s.$$

Its remaining derivative order is $\ell = s - k$, and its scalar is

$$A_r(\alpha, d, k) = \left(\prod_i \binom{\alpha_i}{\beta_i} \binom{\alpha_i - \beta_i}{d_i} \varepsilon_{ri}^{\alpha_i - \beta_i - d_i} \right) [t^k] \prod_i (-\tau_i(t))^{d_i}. \tag{39}$$

Indeed, expand a column first in Z, about $j\omega + \log(1+t)$, then expand each power of $\varepsilon_{ri} - \tau_i(t)$. Extracting $t^s Z^\beta$ gives the stated scalar times the $(\alpha, s-k)$ coefficient of the untranslated column. Indices of weight greater than H have zero coefficient by the column degree bound. This proves the row identity by the binomial theorem and the coefficient formula for a product of power series.

Each scalar satisfies

$$|A_r(\alpha,d,k)| \le \exp\left(-\nu(|\alpha|_w - |\beta|_w) + H\left(T + \frac{\log 2}{w_*}\right)\right). \tag{40}$$

To verify this estimate, let $C_0 = \log(2K) + \nu + \log 2$, so that (34) gives $|\varepsilon_{ri}| \le \exp(C_0 - \nu w_i)$. The binomial factors in (39) have product at most $4^{\sum_i \alpha_i}$. Each logarithmic tail has sum of absolute coefficients at $t = 1/2$ at most $\log 2 < 1$, hence

$$\left|[t^k]\prod_i (-\tau_i(t))^{d_i}\right| \le 2^k.$$

If this coefficient is nonzero, then $\sum_i T_i d_i \le k$. Since $v_0 T_i \ge F_0 w_i$ and $k \le s \le H/v_0$, this gives $|d|_w \le H/F_0$. Finally, $\sum_i \alpha_i \le H/w_*$, and

$$\sum_i (\alpha_i - \beta_i - d_i)(C_0 - \nu w_i) \le \frac{C_0 H}{w_*} - \nu(|\alpha|_w - |\beta|_w) + \frac{\nu H}{F_0}.$$

Combining these bounds proves (40), with exactly the value of T in (17).

More explicitly, multilinearity in the rows gives the finite identity

$$\Delta_\sigma = \sum_f \left(\prod_r A_{r,f}\right) \det B_f, \tag{41}$$

where f chooses one translation term for every row, and B_f has the entries (36). Its orders satisfy $\sum_r \ell_r \le \sum_r s_r \le MH/v_0$, so Proposition 7.1 applies to each term. The number of choices per row is bounded by

$$Q_H = (\lfloor H \rfloor + 1)^{2m}(\lfloor H/v_0 \rfloor + 1). \tag{42}$$

Indeed, $w_i \ge 1$ implies $\alpha_i \le \lfloor H \rfloor$, each d_i has at most $\lfloor H \rfloor + 1$ choices, and $k \le s \le \lfloor H/v_0 \rfloor$. Thus Q_H grows polynomially in H once the parameters have been fixed.

7.3 Two ways for a translation term to be small

Define the low-index count and the collision rate by

$$N_H = \#\{\alpha \in \mathbb{N}^m : |\alpha|_w \le AH\}, \qquad C_H = \frac{c\eta^2 M}{H N_H}. \tag{43}$$

The zero index is included, so $N_H \ge 1$. Fix one nonzero term in (41), and put $W = \sum_r |\alpha_r|_w$. Multiplying (40) over the rows gives the scalar estimate

$$\left|\prod_r A_{r,f}\right| \le \exp\left(-\nu(W - MHb_H) + MH\left(T + \frac{\log 2}{w_*}\right)\right). \tag{44}$$

We have $W \ge MHb_H$.

If $W \ge A(1-\eta)MH$, the first term in the exponent of (44) is at most $-\nu(A(1-\eta) - b_H)MH$. We may discard the nonpositive collision term in (37).

If $W < A(1-\eta)MH$, fewer than $(1-\eta)M$ of the assigned indices can have weight greater than AH. At least ηM assignments therefore fall among the N_H low indices. If their multiplicities are n_α, Cauchy–Schwarz gives

$$\eta^2 M^2 \le \left(\sum_{|\alpha|_w \le AH} n_\alpha \right)^2 \le N_H \sum_{|\alpha|_w \le AH} n_\alpha^2 \le N_H \sum_\alpha n_\alpha^2.$$

The collision saving is consequently at least MHC_H. In this case we may discard the translation saving, since $W - MHb_H \ge 0$.

Each term therefore has either the translation saving or the collision saving. Taking the larger of the two exponential upper bounds and using the at most Q_H^M terms in (41) proves

Proposition 7.2 (Bound at a good embedding)**.** *For every good embedding and every nonzero selected minor,*

$$\frac{\log|\Delta_\sigma|}{MH} \le V + \frac{\log 2}{w_*} + r_a(H) + \max\{-C_H, -\nu(A(1-\eta) - b_H)\}, \tag{45}$$

where

$$r_a(H) = r_{\mathrm{coll}}(M(H), H) + \frac{\log Q_H}{H} \longrightarrow 0.$$

To justify this last limit and the collision limit, the fixed-weight simplex counts give

$$\begin{aligned} M(H) &\sim \frac{K\theta^m}{(m+1)!v_0 \prod_i w_i} H^{m+1}, \\ N_H &\sim \frac{A^m}{m! \prod_i w_i} H^m. \end{aligned} \tag{46}$$

For positive fixed weights these are the usual lattice counts, obtained by rescaling the simplex and comparing the lattice cubes along its boundary. Strict versus non-strict inclusion changes only the lower-order boundary contribution. In particular $\log M(H)/H \to 0$, and (42) gives $\log Q_H/H \to 0$. Dividing the two asymptotics in (46) also gives

$$C_H \longrightarrow \frac{c\eta^2 K\theta^m}{(m+1)v_0 A^m} = C_\infty. \tag{47}$$

These limits are taken with the dimension and approximation weights fixed.

8 The remaining embeddings and the multiplier cost

At any embedding, including one outside G, (14) gives $|p_i/q_i| \le |x| + 1$. Hence

$$|r_{\sigma i}| \le |\sigma(u)|(|x| + 1 + |\sigma(v)|) \le R. \tag{48}$$

An entrywise coefficient estimate is enough here. On the circles $|t| = 1/2$, $|Z_i| = 1$, we have

$$|L_i(t)| \le \sum_{k \ge 1} \frac{2^{-k}}{k} \le 1, \quad |1+t| \le 3/2, \quad |Z_i + jr_{\sigma i} + L_i(t)| \le KR + 2.$$

Cauchy's coefficient estimate therefore bounds an entry in (25) by

$$2^s \exp(\kappa K h)(3/2)^h (KR+2)^{\sum_i \gamma_i}.$$

The row and column budgets give $s \le H/v_0$, $h \le H/w_0$, and $\sum_i \gamma_i \le H/w_*$. Increasing $KR+2$ to $2(KR+2)$ makes this at most $\exp(HU(R))$, with U from (17). Expanding the determinant into $M!$ permutation terms yields the following bound.

Proposition 8.1 (Coarse determinant bound)**.** *At every embedding, for a nonzero selected minor,*

$$\frac{\log|\Delta_\sigma|}{MH} \le U(R) + r_c(H), \qquad r_c(H) = \frac{\log M(H)}{H} \longrightarrow 0. \tag{49}$$

Indeed $\log(M!)/(MH) \le \log M/H$, and the limit follows from (46). The bound is uniform in the chosen columns and in the embedding; it does not need $e^\omega = z_\sigma$.

There remains the extra $\log 2/w_*$ in the good estimate. It is small, but fixed as H grows, so it cannot be included in $r_a(H)$. Instead, for $K \ge 1$ and $R \ge 0$,

$$2 \cdot 2(KR+2) \le 2(K(2R+2)+2).$$

Taking logarithms gives

$$U(R) + \frac{\log 2}{w_*} \le U(2R+2) = U(R'). \tag{50}$$

This is why the parameter construction used R' from the outset.

9 The determinant contradiction

We can now complete the proof of Theorem 3.1. For a height at which interpolation supplies a nonzero selected minor, apply Proposition 7.2 at the two good embeddings and Proposition 8.1 at the other two. Lemma 6.3 and (50) give

$$-2(1-b_H) - 2E - U(R') - r_c(H) \le V + r_a(H) + \max\{-C_H, -\nu(A(1-\eta) - b_H)\}. \tag{51}$$

Put

$$S_H = 2E + U(R') + V + r_a(H) + r_c(H). \tag{52}$$

The strict margins (19), the limits of the two remainders, and (47) imply that, for all sufficiently large H,

$$S_H < g, \qquad 2 + S_H < C_H. \tag{53}$$

Suppose first that the maximum in (51) is the approximation term. Rearranging gives

$$\nu(A(1-\eta) - b_H) - 2(1-b_H) \le S_H < g.$$

But, using (8), its left side is

$$g + (\nu - 2)(\theta - b_H) \ge g,$$

because $\nu > 4$ and $b_H \le \theta$. This is a contradiction.

Suppose instead that the maximum is the collision term. Then (51) gives

$$C_H \le 2(1-b_H) + S_H \le 2 + S_H,$$

contradicting the second inequality in (53). If the two terms are equal, either argument applies.

We have shown that full row rank is impossible at every sufficiently large height. Lemma 5.2 supplies full row rank at a cofinal set of heights, which is incompatible with that conclusion. Thus (2) is false for the fixed $\nu > 4$. The logical reduction in Section 2 proves (1) for every $\nu > 4$, and Lemma 2.1 proves irrationality and the exponent bound. This completes the proof of Theorem 3.1.

10 The three concrete constructions

Use the field

$$F = \mathbb{Q}(\sqrt{3})(\sqrt{2}) = \mathbb{Q}(\sqrt{2}, \sqrt{3}). \tag{54}$$

Both generators are algebraic integers. For completeness, its degree is four: $\mathbb{Q}(\sqrt{3})$ has degree two, and if $(r + s\sqrt{3})^2 = 2$ with $r, s \in \mathbb{Q}$, comparison of coefficients gives $2rs = 0$. The case $s = 0$ would make two a rational square. The case $r = 0$ gives $3s^2 = 2$, hence $(3s)^2 = 6$, making six a rational square. Both are impossible. Adjoining $\sqrt{2}$ therefore doubles the degree.

Its four embeddings are the real maps

$$\sigma_{\varepsilon,\tau}(\sqrt{2}) = \varepsilon\sqrt{2}, \qquad \sigma_{\varepsilon,\tau}(\sqrt{3}) = \tau\sqrt{3}, \qquad \varepsilon, \tau \in \{+1, -1\}. \tag{55}$$

Each application uses the distinguished embedding $\sigma_{+,+}$. The data and good embeddings are summarized below; we verify the compatibility identities individually.

Target	a	u	v	Good embeddings
x_1	2	1	$\sqrt{2}$	$\sigma_{+,+}, \sigma_{+,-}$
x_2	4	$\sqrt{2}$	0	$\sigma_{+,+}, \sigma_{+,-}$
x_3	$3 + 2\sqrt{2}$	$\sqrt{2}$	$\sqrt{3}$	$\sigma_{+,+}, \sigma_{-,+}$

10.1 The algebraic shift

Take $a = 2$, $u = 1$, $v = \sqrt{2}$, and $x = x_1 = \log 2 + \sqrt{2}$. These elements are integral, $x_1 > 0$, and the distinguished value of a is $2 > 0$, different from one. At either good embedding,

$$\sigma(u)(x_1 - \sigma(v)) = \log 2, \qquad \exp(\log 2) = 2 = \sigma(a).$$

Also $|\sigma(u)| = 1 \le 2$. The two good embeddings are distinct because they have different values on $\sqrt{3}$. All hypotheses of Theorem 3.1 hold, proving (1) for x_1 and hence Theorem A.

10.2 The algebraic multiple

Take $a = 4$, $u = \sqrt{2}$, $v = 0$, and $x = x_2 = \sqrt{2}\log 2$. Again all the algebraic data are integral, the target is positive, and the distinguished base is positive and not one. At either good embedding,

$$\sigma(u)(x_2 - \sigma(v)) = \sqrt{2}\,\sqrt{2}\log 2 = 2\log 2, \qquad \exp(2\log 2) = 4 = \sigma(a).$$

The multiplier bound is $|\sigma(u)| = \sqrt{2} \le 2$. Theorem 3.1 gives (1) for x_2, and Lemma 2.1 gives Theorem B.

In these first two constructions the algebraic data lie in the smaller field $\mathbb{Q}(\sqrt{2})$. The auxiliary $\sqrt{3}$ supplies two distinct embeddings with the required values. Each is a distinct embedding of the degree-four field and contributes a factor to the norm.

10.3 The shifted quadratic period

Put $b = 3+2\sqrt{2} > 1$, and take $a = 3+2\sqrt{2}$, $u = \sqrt{2}$, $v = \sqrt{3}$. The target is $x_3 = \log b/\sqrt{2}+\sqrt{3} > 0$. At $\sigma_{+,+}$,

$$\sigma_{+,+}(u)(x_3 - \sigma_{+,+}(v)) = \log b,$$

so the exponential is $b = \sigma_{+,+}(a)$. At $\sigma_{-,+}$, the same computation gives $-\log b$. Now

$$(3 + 2\sqrt{2})(3 - 2\sqrt{2}) = 1, \tag{56}$$

so

$$\exp(-\log b) = b^{-1} = 3 - 2\sqrt{2} = \sigma_{-,+}(a).$$

Both good embeddings have multiplier absolute value $\sqrt{2} \leq 2$, and all three field elements are integral. Theorem 3.1 therefore proves (1) for x_3, and proves Theorem C.

Here the two good embeddings change the sign of the normalized logarithmic coordinate while keeping the shift $\sqrt{3}$ fixed. Identity (56) is what makes their exponential bases match. No compatibility is needed at the embeddings that change the sign of the shift; their contribution has already been covered by the coarse bound.

11 Further questions

The exact values of the three irrationality exponents remain undetermined by this argument: we obtain $2 \leq \mu(x_j) \leq 4$. The proof gives no numerical value for the threshold $Q(x, \nu)$ in (1). Can the parameter choices be made effective enough to give an explicit threshold?

A stronger question at exponent four is whether, for each of the three numbers x, there exists $c_x > 0$ such that

$$\left| x - \frac{p}{q} \right| \geq c_x q^{-4}$$

for every integer p and all sufficiently large positive integers q. The bound $\mu(x) \leq 4$ does not imply this estimate.

Here two compatible embeddings supply analytic savings, while the norm has four factors. What exponent bound can be obtained in a field of degree d with k compatible embeddings? A related question is how the bound changes when the affine data are not integral and require additional denominator clearing.

A Formalization and verification

The proof is formalized in Lean in the accompanying `formalizations/AffineLogarithms` package [6]. Theorem 3.1 is formalized as

`OAI.PairedAffine.Endpoint.paired_eventualLowerBound`.

Its parameters are exactly the field elements, integrality proofs, degree, distinguished embedding, real target, and good-embedding conditions in Theorem 3.1.

A.1 Statement correspondence

For each of the three targets, the formalization proves the eventual bound (1), irrationality, and the exponent upper bound. It also derives the exclusion of Mathlib's `LiouvilleWith` $\nu\, x$ for every $\nu > 4$. In the namespace `OAI.PairedAffine`, the declaration stems are

Theorem	Declaration stem
A	`log_two_add_sqrt_two`
B	`sqrt_two_mul_log_two`
C	`shifted_period`

The corresponding suffixes are `_eventualLowerBound`, `_irrational`, and `_irrationalityExponent_le_four`. The package records the full statements and their correspondence with Theorems A–C.

The exponent convention is in `OAI/NumberTheory/PiExponent/Statement.lean`; its use in this package is justified by `AlgebraicLog/ExponentConsequence.lean`. The Mathlib formulations are in `Solution.lean`; `StandardBridge.lean` supplies the implication from the eventual bound to the exclusion of `LiouvilleWith`.

A.2 Where the main steps are checked

The following table locates the main parts of the formal proof. Paths are relative to the package root; declaration names below omit the initial `OAI.`.

Mathematical step	Source and declaration
Shape and parameter margins	`AlgebraicLog/ParameterShape.lean`, `exists_quadratic_shape`; `AlgebraicLog/DimensionBudget.lean`; `AlgebraicLog/AdmissibleParameters.lean`, `exists_admissible_parameters`, `error_sum_lt_gap`, and `collision_exceeds_error_sum`.
Weighted interpolation	`Logarithm/Interpolation.lean`, `eventually_weighted_logarithmic_interpolation`; its hypotheses are the fields of `Logarithm/GeometryData.lean`.
Truncation and full row rank	`Logarithm/ScaledTruncation.lean`, `formalLog_packets_surjective_iff_truncated`; `Logarithm/ScaledMatrix.lean`, `truncatedLogMatrix_surjective_of_formalLog_packets`; `PairedAffine/Geometry.lean`, `cofinal_matrix`.
Integral entries and selected minor	`OAI/NumberTheory/PiExponent/Approximation/Arithmetic.lean`, `log_lcmUpto_le`; `PairedAffine/ArithmeticClearing.lean`, `exists_integral_entry_lift`; `PairedAffine/Arithmetic.lean`, `exists_integral_selected_minor` and `log_selectedScalar_le`.
Norm comparison	`PairedAffine/NormSelected.lean`, `paired_selected_normalized_bound`.
Collision estimate	`Logarithm/ScaledCollision.lean`, `formal_scaled_collision_bound`; the analytic foundation is in `OAI/NumberTheory/PiExponent/Analysis/`.
Finite translation and good bound	`PairedAffine/Analytic.lean`, `norm_choiceScalar_le`, `minor_expansion`, and `minor_analytic_bound`.
Coarse bound and multiplier	`PairedAffine/AnalyticCoarse.lean`, `minor_log_bound` and `conjugateCost_add_log_two_le`.

Mathematical step	Source and declaration
Contradiction and endpoint	`AlgebraicLog/ContradictionNumerics.lean`, `quadratic_determinant_bounds_inconsistent`; `PairedAffine/Endpoint.lean`, `no_fixed_data` and `paired_eventualLowerBound`.
Concrete field and targets	`PairedAffine/Field.lean`, `PairedAffine/AffineExamples.lean`, `PairedAffine/Concrete.lean`, and `PairedAffine/Main.lean`.

A.3 Verification

The formalization uses Lean 4.34.1 and Mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612`. Its permitted foundational axioms are `propext`, `Classical.choice`, and `Quot.sound`. The statements described above passed exact statement comparison and proof checking with Lean and Rust Nanoda, an independently implemented kernel.

The checked source version is identified in Appendix B. The accompanying verification guide gives reproduction instructions and records the checker versions, dependency caches, and trust assumptions.

B Source availability and attribution

The proof package builds on repository version `0cfe10002ea95c45721b93d18bffbe2c3bbbcbbd` in the `Mattie/math` repository. It incorporates OpenAI's `OAI.NumberTheory.PiExponent` library at revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The rational-logarithm and quadratic-logarithm extensions supply mathematical constructions and estimates used throughout this paper. The original affine work extends that framework to the common integral affine data and the paired embedding argument.

The rational-logarithm sources in that repository version are in `preprints/The-irrationality-exponent-of-positive-rational-logarithms-is-2-October-7-2026/lean`. The quadratic-logarithm modules used here are included under `AlgebraicLog` in the affine package. The checked affine sources are in `formalizations/AffineLogarithms` at repository version `b15a6431eca8632a48c1cd84e5200edd30f0670c`; the package records the exact source files and their hashes. Appendix A describes the verification of this version.

Licenses and attribution notices accompany the source package.

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